

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 5: ACTIVE CONTOURS

SLO 2: Implement and evaluate active contour models for segmenting objects in binary and grayscale images.

SNAKE PATH FINDER SOLUTIONS

Binary Grid (Row, Column Indexing):

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	1	1	1	0
2	0	0	1	1	1	0
3	0	0	1	1	1	0
4	0	0	0	0	0	0

Starting Position: (1, 1) | Object Boundary Begins at Column Index 2.

1. Direction to move to reach the object boundary:

Right (or East / Rightwards along the row towards Column 2).

2. Coordinates of the first three positions the snake will move to:

Step 1: (1, 2) — Reaches the upper-left corner boundary of the object.

Step 2: (1, 3) — Tracks along the top horizontal boundary pixel.

Step 3: (1, 4) — Tracks to the upper-right corner boundary pixel.

(Alternative valid tracking sequence along the vertical edge: Step 1: (1, 2), Step 2: (2, 2), Step 3: (3, 2)).

ENERGY MINIMIZATION SOLUTIONS

Unscramble the Energy Types:

1. NITRAELN YRENEG →

INTERNAL ENERGY (Controls contour elasticity and bending smoothness).

2. AXRETENL YRENEG →

EXTERNAL ENERGY (Attracts contour toward key image features like edges).

3. EGDYEAB YRENEG →

IMAGE ENERGY (Represents image-based forces derived from intensity gradient).